



Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats

CLOUDFLARE RADAR · ANNUAL REPORT 2024

17% Global Traffic Growth and Starlink's 3.3x Surge

Global Internet traffic accelerated 17% YoY; satellite connectivity is reshaping access patterns in underserved regions


17%

Global Internet traffic growth YoY


3.3x

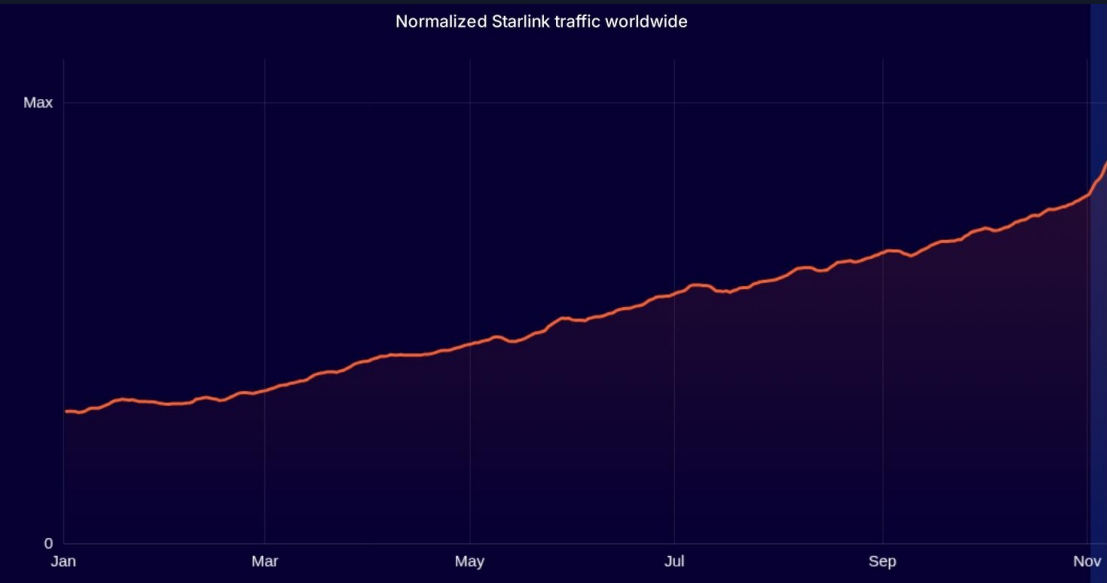
Starlink traffic surge in 2024


41%

Global traffic from mobile devices


33%

Mobile traffic from iOS devices



Normalized Starlink traffic worldwide showing 3.3x growth in 2024

Trend scaled relative to peak observed request volume. Starlink not yet available globally.

Key Insights

- Growth driven by increasing reliance on Internet for communication, commerce, entertainment, transportation
- Starlink brings satellite Internet to previously underserved locations, expanding global connectivity footprint
- Android peaks at >95% traffic share in some regions; iOS peaks around 66%
- Spain recorded the highest download speed among all countries measured

Source: Cloudflare Radar 2024 Year in Review

Google Leads Top 10 Services as Generative AI Rises

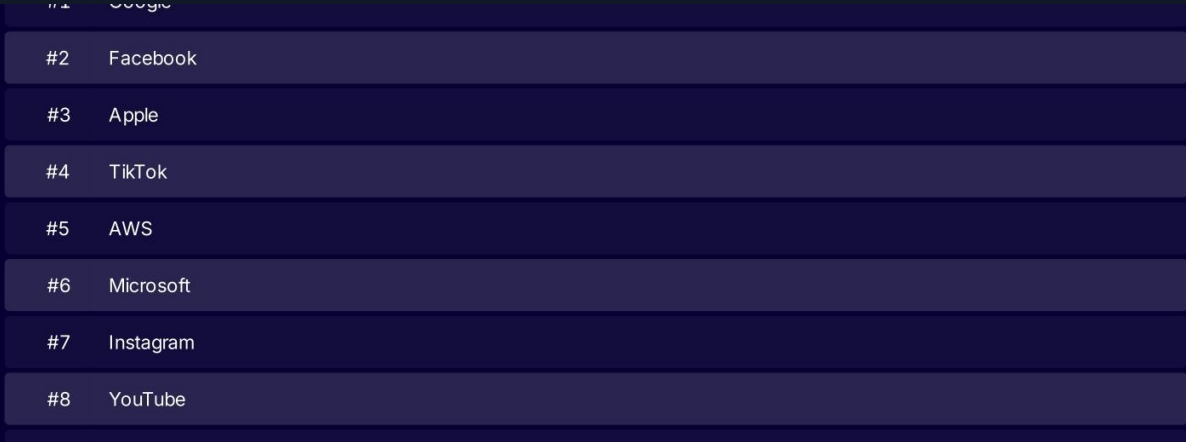
Google remains #1 for the second consecutive year; generative AI rises as the Metaverse fades

Top 10 Most Popular Internet Services

Based on aggregate data from Cloudflare 1.1.1.1 DNS resolver

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Google maintained #1 for the second consecutive year



Top 10 most popular Internet services worldwide



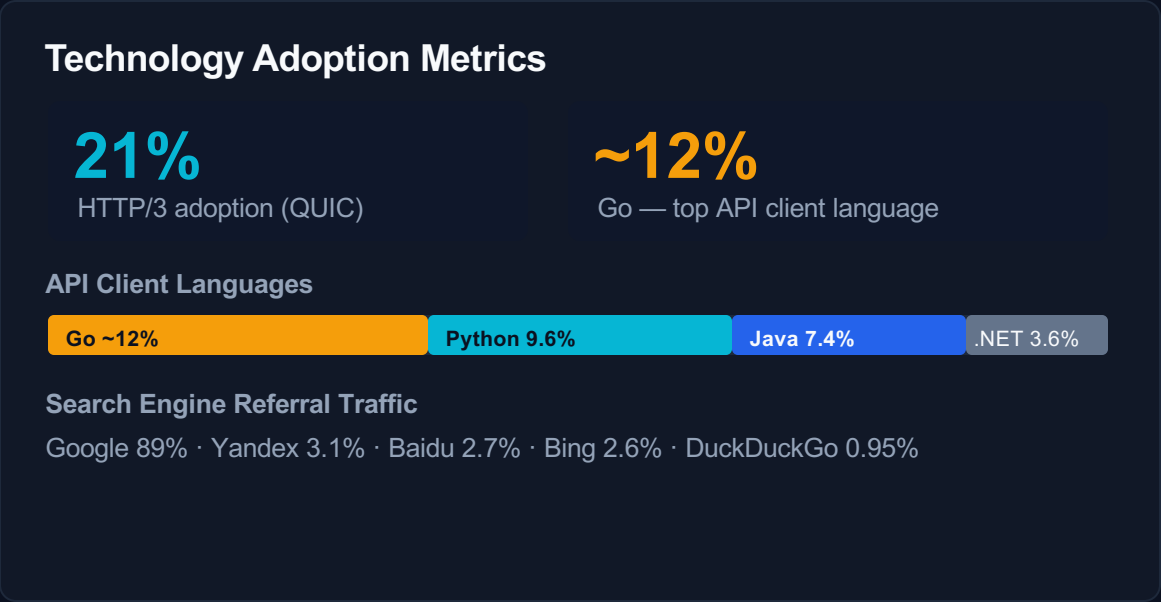
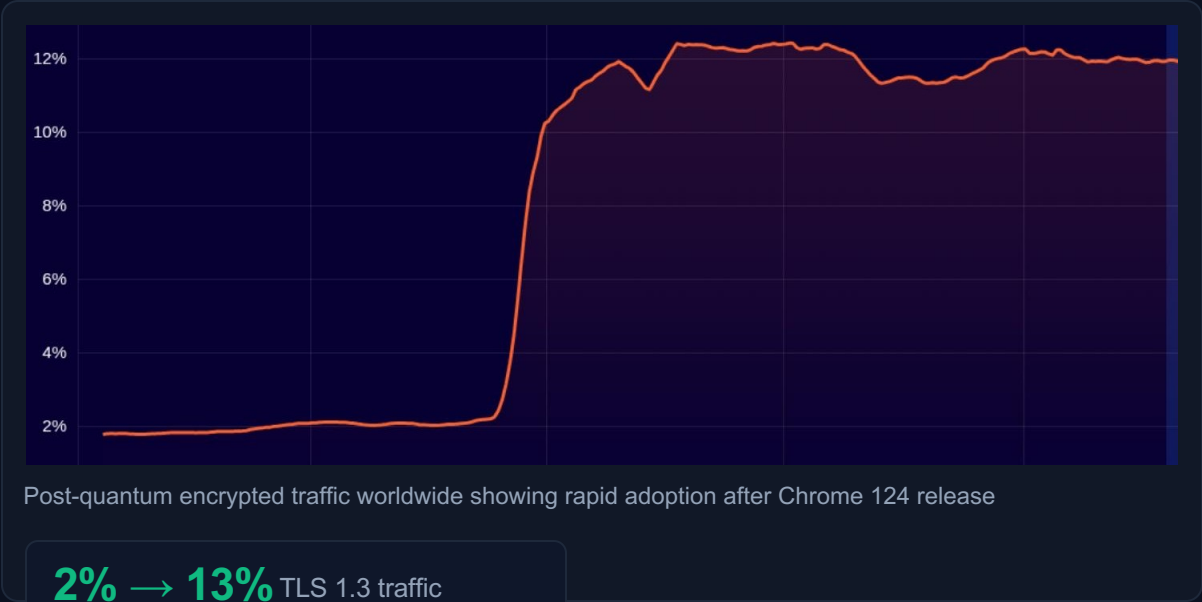
Generative AI on the Rise

- Generative AI continued its rapid rise throughout 2024, cementing itself as a dominant force in Internet services
- The Metaverse receded into the background as practical AI applications captured user and enterprise attention

Source: Cloudflare Radar 2024 · Ranking via 1.1.1.1 DNS resolver data

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Post-quantum encryption saw dramatic adoption; HTTP/3 and Go also gained ground across the stack



Post-Quantum Encryption Surge

- Chrome 124 enabled post-quantum key agreement by default on April 17, 2024 — triggering the sharp climb
- Mozilla Firefox rolling out by default; Apple Safari began initial testing in 2024

Developer Ecosystem Shift

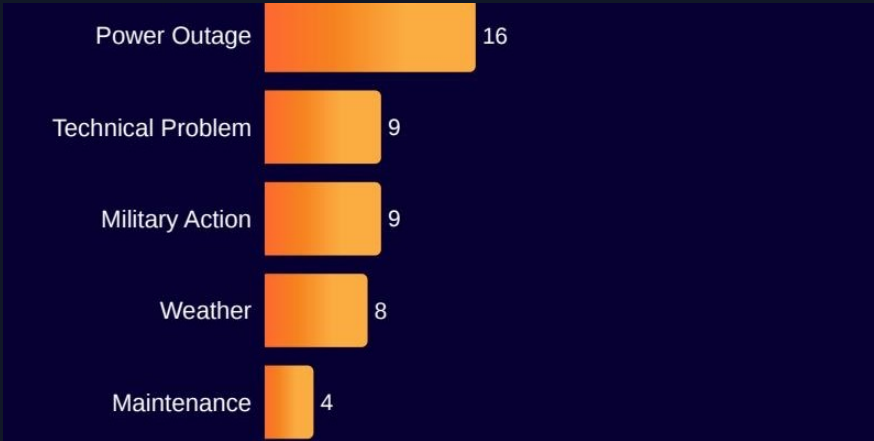
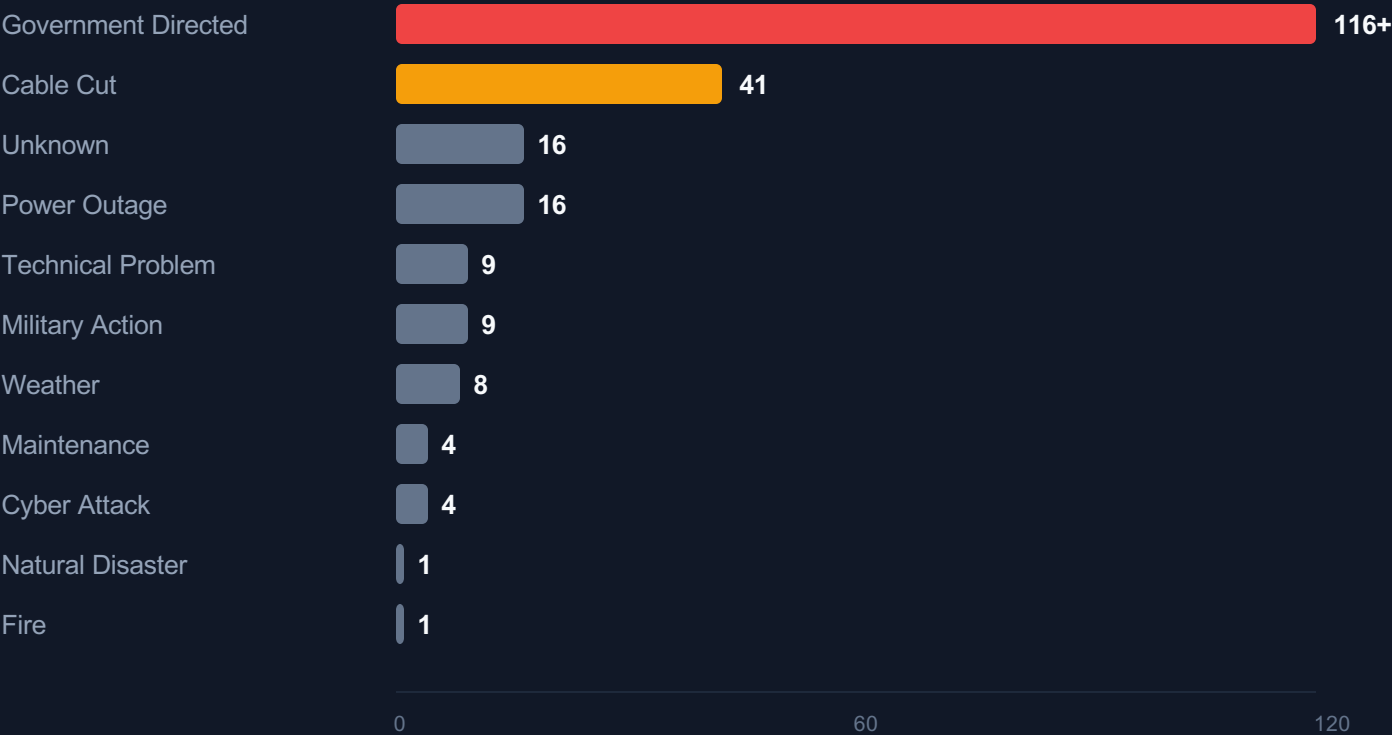
- Go surpassed other languages to become the most popular API client language, capturing ~12% of automated API traffic
- HTTP/3 on QUIC delivers faster connections, packet-loss mitigation, and encryption by default

225 Major Disruptions: Government Shutdowns Lead Outage Causes

225 major disruptions observed globally; government-directed shutdowns are the leading cause by a wide margin

Causes of Major Internet Disruptions (2024)

Total: 225 major disruptions observed via Cloudflare Radar Outage Center



Causes of major Internet disruptions observed in 2024

Additional Connectivity Metrics

28%

IPv6 dual-stack traffic

41%

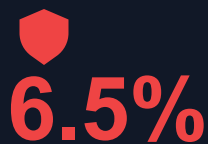
Traffic from mobile

- Spain recorded the highest download speed among all countries measured

Source: Cloudflare Radar Outage Center

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Security threats remain pervasive; mitigation, bot concentration, and legacy vulnerabilities persist



Of all traffic mitigated



Bot traffic from top 10 countries



Emails were malicious



RPKI valid routes increase

Persistent Threats

- **Gambling and Games** was the most attacked industry category across Cloudflare's network
- **Log4j (CVE-2021-44228)** — first disclosed December 2021 — continues to be actively targeted by attackers more than two years after disclosure
- This underscores the persistent nature of widely known vulnerabilities in production infrastructure

Source: Cloudflare Security Analytics 2024

Email Security Threats



Of emails processed were malicious



Of malicious emails had deceptive links

- The **.bart** top-level domain originated the largest share of malicious and spam email observed
- Deceptive links remain the primary delivery mechanism for email-based threats

Source: Cloudflare Email Security Analysis 2024

Five Takeaways Shaping the 2025 Internet Outlook

Five strategic findings to inform 2025 planning for network and security leaders



Traffic Growth Trajectory

17% YoY growth + Starlink 3.3x surge signals expanding global connectivity; plan infrastructure for sustained demand



Post-Quantum Momentum

2% → 13% leap in TLS 1.3 means crypto-agility is now a board-level priority; audit encryption roadmaps now



Connectivity Fragility

Government-directed shutdowns (116+) remain the #1 disruption cause; diversify paths and build resilience



Persistent Security Threats

6.5% mitigation rate, Log4j still exploited, email threats at 4.3% — defense-in-depth and patching discipline are non-negotiable



Evolving Developer Ecosystem

Go leads API clients at ~12%; HTTP/3 at 21% reshaping protocols — align architecture with modern stack trends